Numerical Oscillators: Exercise Guide for 2A

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Mathematical Methods for Physicists II

Coursework Notes and Computational Guides

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This is a collection of notes from my Mathematical Methods for Physicists II coursework on numerical oscillator experiments. It accompanies notebook 2A. The shared foundations guide introduces the theory; this guide explains how each exercise translates that theory into a computational procedure.

1 How to use this guide

Read the shared foundations first, particularly the coupled-state formulation and the numerical-method definitions. This guide follows the revised 2A notebook. The goal is to reconstruct each algorithm from its equations, rather than copy code without understanding its update order.

The notebook's original scenarios come from homework, quiz, and assignment material. The revised comparisons use common physical durations and add full-trajectory convergence diagnostics. These changes affect the experimental design, not the oscillator equations.

2 Exercise 1: harmonic motion with Euler and midpoint RK2

2.1 Define the problem before writing the loop

The model and initial conditions are

$$\dot{x} = v, \quad \dot{v} = -5x, \qquad x(0) = 1, \quad v(0) = 0.$$

Its exact solution is $x(t) = \cos(\sqrt{5}t)$ and $v(t) = -\sqrt{5}\sin(\sqrt{5}t)$. Both variables must be evolved, even when the requested plot shows only position.

The first Euler comparison uses $h = 0.02, 0.08, 0.8, 1$ and a common final time $T = 20$. The step counts are respectively 1000, 250, 25, and 20. The second comparison uses RK2 with $h = 0.08, 0.05$ and Euler with $h = 0.05$, all ending at $T = 10$; their step counts are 125, 200, and 200.

The two panels of comparisons therefore have different stated durations, but all runs *within each comparison* share their duration. Retaining 200 steps for every h would change both resolution and total evolution time.

2.2 Construct a reusable solver

Choose arguments for the method, h, number of steps, c, and initial state. Initialize current variables x, v and histories containing their initial values. At each iteration:

1. Calculate the method's proposed new position and velocity from the current state.
2. Commit both new values after completing the calculation.
3. Append $t = (i + 1)h$ and the two new components.

Return all three histories. The loop index i begins at zero; the newly calculated state corresponds to $(i+1)h$.

For Euler use

$$x_{\text{new}} = x + hv, \qquad v_{\text{new}} = v - hcx.$$

For midpoint RK2 first calculate

$$x_{\text{mid}} = x + hv/2, \quad v_{\text{mid}} = v - hcx/2,$$

then use $x_{\text{new}} = x + hv_{\text{mid}}$ and $v_{\text{new}} = v - hcx_{\text{mid}}$. Do not replace the original x by the midpoint before computing all the needed quantities.

2.3 Build and interpret the plots

Evaluate the exact solution at each run's own time array. A separate subplot for each Euler step prevents a very large unstable amplitude from hiding the other curves. The RK2/Euler overlay uses the common interval $0 \le t \le 10$ and a limited vertical range; inspect the energy plot later for behaviour outside that display range.

Forward Euler introduces energy gain: $e_{n+1} = (1 + 5h^2)e_n$. A large numerical amplitude is not a physical instability of the spring. Midpoint RK2 has smaller errors at these small steps, but it also has artificial harmonic-oscillator energy gain. The exact reference distinguishes numerical growth from the true constant-amplitude motion.

2.4 Checks

Each history should have $N+1$ entries, begin at $(0, 1, 0)$, and end at the intended T. Compare the initial acceleration with -5. Check that Euler's two updates use the same old state and RK2's two slopes use the same midpoint state.

3 Exercise 2: compare methods and test convergence

3.1 Trajectory and energy at the same resolution

Use $h = 0.05$, 200 steps, and $T = 10$ for Euler, midpoint RK2, and RK4. The physical problem is identical for all three; only the update method changes.

RK4 evaluates four pairs of slopes. For example, $k_{1x} = v$ and $k_{1v} = -5x$. Stage 2 uses $(x + hk_{1x}/2, v + hk_{1v}/2)$, stage 3 uses the analogous pair from stage 2, and stage 4 uses a full-step prediction from stage 3. Combine each component with weights $1, 2, 2, 1$ divided by six.

For every saved state calculate

$$e_n = \tfrac{1}{2}(v_n^2 + 5x_n^2), \qquad e_0 = 2.5.$$

This is energy per unit mass. Plot the analytic constant alongside the numerical histories. Euler gains substantial energy, RK2 gains less, and RK4 is much more accurate on this interval. These are observations for the specified problem and step, not a universal ranking for every long-time simulation.

3.2 Why an endpoint error is insufficient

An oscillatory numerical solution can have incorrect amplitude and phase yet happen to be close to the reference at the final instant. The revised notebook therefore reports both

$$\epsilon_T = |x_h(T) - x_{\text{exact}}(T)|$$

and

$$\epsilon_\infty = \max_n |x_h(t_n) - x_{\text{exact}}(t_n)|.$$

The maximum is taken over the saved grid points, not all real times. Refining that grid still makes it a useful full-interval diagnostic.

3.3 Reconstruct the refinement algorithm

The step sizes are $0.2, 0.1, 0.05, 0.025, 0.0125$, always to $T = 10$.

1. For each method, start an empty error list and no previous error.
2. For each h, set $N = \text{round}(10/h)$ and integrate.
3. Evaluate the exact position at every saved time.
4. Subtract, take absolute values, and calculate the endpoint and maximum errors.
5. After the first run, calculate $p_{\text{obs}} = \log_2(\epsilon_{\text{previous}}/\epsilon_{\text{current}})$.
6. Store the maximum error and draw a log–log error-versus-step plot.

The first observed order is undefined and is printed as NaN. When $\epsilon \approx Ch^p$, halving h divides error by 2^p, so the displayed order approaches p.

3.4 What the revised outputs show

RK2 approaches order two and RK4 order four. At the two finest displayed resolutions their maximum position errors give observed orders about 2.001 and 4.002, respectively. Euler's order approaches one more slowly; the displayed finest pair gives about 1.24. Its coarse runs are strongly amplified and not yet in the clean asymptotic regime.

One RK4 endpoint error increases when h changes from 0.2 to 0.1 because endpoint cancellation changes, while its maximum trajectory error decreases. This is precisely why the new diagnostic is useful. These numbers are checks for the revised run, not constants of the methods.

4 Exercise 3: power-law forces with RK4

4.1 Define the new concepts

A nonlinear force depends nonlinearly on position. Nonlinear does not automatically mean unstable or chaotic. Here

$$\dot{x} = v, \qquad \dot{v} = -x^n, \qquad n = 1, 2, 3.$$

The scaled potential is $u = x^{n+1}/(n+1)$ and energy per unit mass in these variables is $e = v^2/2 + u$. Odd n gives a restoring force on both sides of zero; $n = 2$ does not.

All cases begin at $(x, v) = (1, 0)$ with $h = 0.01$, up to 1000 steps ($T = 10$ unless stopped). The $n = 1$ case now has frequency one, not $\sqrt{5}$ as in exercises 1 and 2.

4.2 Construct the nonlinear stages

Use the same RK4 structure as before, but evaluate $-x^n$ at each stage's own position. For instance,

$$x_2 = x + hk_{1x}/2, \quad v_2 = v + hk_{1v}/2, \quad k_{2x} = v_2, \quad k_{2v} = -x_2^n.$$

Repeat with the stage-2 pair for stage 3, and a full stage-3 prediction for stage 4. Advance both components with the weighted slopes. Store histories and plot position and velocity in separate panels for each power.

4.3 Understand the cutoff and the physics

For $n = 2$, acceleration remains negative after x becomes negative. The potential has no lower bound there, allowing runaway motion. The notebook stores the newly calculated state and stops once $|x| > 50$. Consequently the final stored point can already lie beyond 50; the cutoff is not an exactly located event.

This avoids further overflow. It neither reflects the trajectory nor clips its position. The force law explains why runaway is plausible, but the rapidly accelerating tail still needs smaller-step checks for quantitative accuracy. Do not interpret a safeguard alone as validation of the computed escape time.

For $n = 3$, the quartic well confines finite-energy motion; the oscillation is nonlinear and need not be sinusoidal. The $n = 1$ case supplies an additional simple exact check, $x = \cos t$.

5 Exercise 4: a physical spring with velocity Verlet

5.1 Parameters and analytic target

The spring has $k = 5$, $m = 15$, equilibrium zero, $x_0 = 0.1$, and $v_0 = 0.1$. Use $h = 0.01$ for 5000 steps, hence $T = 50$. Interpreting these as SI values, k is in N/m, m in kg, x in m, and v in m/s.

$$\Omega_0 = \sqrt{5/15}, \qquad x_{\text{exact}} = 0.1\cos(\Omega_0 t) + \frac{0.1}{\Omega_0}\sin(\Omega_0 t).$$

The nonzero initial velocity requires the sine term.

5.2 Build the force and the update

Define $F(x) = -k(x - x_{\text{eq}})$. Initialize histories for position, velocity, force, and time. Each step uses the last saved values:

1. Calculate $x_{\text{new}} = x + hv + h^2F/(2m)$.
2. Calculate F_{new} at the new position.

3. Calculate $v_{\text{new}} = v + h(F + F_{\text{new}})/(2m)$.

4. Append all new values and the new time.

Forces must be divided by mass. The velocity update averages accelerations, not just velocities or positions.

5.3 Energy and accuracy checks

Now calculate *total* mechanical energy,

$$E = \tfrac{1}{2}(15)v^2 + \tfrac{1}{2}(5)x^2, \qquad E_0 = 0.1\ mJ.$$

The notebook plots this energy and prints the maximum position difference from the analytic curve and the relative energy range,

$$\frac{\max E - \min E}{E_0}.$$

A narrow energy range is characteristic of a well-resolved stable Verlet simulation. It is not exact conservation and does not rule out accumulated phase error. Here $h\Omega_0 \approx 0.00577$ is comfortably smaller than the harmonic stability threshold two.

6 Writing and interpreting your own version

Separate the physical model from its integration method: a force function supplies acceleration, a step function supplies the update rule, and a loop supplies histories. The essential checks are initial conditions, common duration, exact references where available, consistent energy definitions, and step-size refinement.

Preserve the distinction between a numerical instability in a stable harmonic oscillator and physical runaway in the $-x^2$ model. When modifying parameters, reconsider the fastest timescale and the force's potential before reusing a previously adequate step size.